

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 5: Acids, Bases and Salts

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	5 — Acids, Bases and Salts
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Paper 5 (hardest tier). Expert traps: reading a chain of indicator results where some indicators (phenolphthalein, turmeric) cannot separate acid from neutral and only a second indicator settles the family; tracking a drop-by-drop neutralisation past the endpoint into overshoot, then reasoning what every indicator shows afterwards; choosing the correct member of an acid-base pair to fix a sting, soil, effluent, antacid or pond when the scenario is layered or reversed; never confusing the colour an indicator displays with the family it names (china rose magenta = acid, turmeric red = base, methyl orange red = acid, phenolphthalein pink = base); knowing that the salt of a neutralisation depends on the particular acid and base and is rarely common salt; and never letting strength or amount of sourness flip a substance out of its chemical family. Do not write on this paper — mark answers on your answer sheet.

1. Four colourless liquids are dilute hydrochloric acid, sugar water, sodium hydroxide solution and common salt water. A single strip of red litmus is dipped into each. Counting how many strips turn blue gives:

- (A) exactly two, the sodium hydroxide solution and the salt water
- (B) exactly three, all except the hydrochloric acid
- (C) exactly one, the sodium hydroxide solution
- (D) none, because red litmus only ever turns red

2. A liquid leaves both blue and red litmus unchanged, leaves phenolphthalein colourless, and leaves turmeric yellow. A student announces this proves the liquid is distilled water. The most accurate verdict is:

(A) the student is right, because only distilled water leaves all four indicators unchanged

(B) the student is wrong, because distilled water would have turned phenolphthalein faintly pink

(C) the student is wrong, because distilled water turns turmeric red

(D) the tests only establish the liquid is neutral; sugar water and common salt water would behave identically, so it need not be distilled water

3. Methyl orange is red in acids and yellow in bases. A drop is added to fresh samples of vinegar, sodium hydroxide solution and sugar water in turn. The colours seen are, respectively:

(A) red, yellow, red

(B) red, yellow, yellow

(C) yellow, red, yellow

(D) red, red, yellow

4. A pink, phenolphthalein-treated lime water is titrated by adding lemon juice drop by drop. The pink just vanishes after about eighteen drops. The student then adds eight more drops of lemon juice and tests with blue litmus. The blue litmus now:

(A) stays blue, because the mixture is locked at neutral once the pink disappears

(B) turns red, because the extra acid has pushed the mixture past neutral into the acidic range

(C) turns red, but the phenolphthalein also turns pink again from the excess acid

(D) stays blue, because lime water can never become acidic

5. Dilute nitric acid is exactly neutralised by potassium hydroxide; separately, dilute hydrochloric acid is exactly neutralised by sodium hydroxide. After each mixture is evaporated, the solids are:

(A) common salt (sodium chloride) in both cases

(B) the leftover bases, since some base always survives

(C) two different salts, potassium nitrate from the first and sodium chloride from the second

(D) no solid at all, because neutralisation destroys both substances

6. A child is stung by a wasp (whose venom is alkaline) and, separately, by a red ant (which injects an acid). The correct first-aid for the two stings is, respectively:

(A) mild vinegar for the wasp sting and baking soda paste for the ant sting

(B) baking soda paste for the wasp sting and mild vinegar for the ant sting

(C) baking soda paste for both stings (D) mild vinegar for both stings

7. Three liquids are tested. P turns red litmus blue. Q turns blue litmus red. R leaves both litmus papers unchanged. Each is then given the china rose indicator (magenta in acids, green in bases). The liquid that turns china rose green is:

(A) Q

(B) R

(C) P

(D) both P and R

8. A solution turns china rose magenta. Predicting how the same solution behaves with red litmus, with phenolphthalein and to the touch gives:

- (A) red litmus stays red, phenolphthalein stays colourless, and it tastes sour
- (B) red litmus turns blue, phenolphthalein turns pink, and it feels soapy
- (C) red litmus stays red, phenolphthalein turns pink, and it tastes sour
- (D) red litmus turns blue, phenolphthalein stays colourless, and it feels soapy

9. A bottle's contents turn blue litmus red, leave phenolphthalein colourless, turn turmeric yellow and turn china rose magenta. Reading all four results together, the contents are:

- (A) a base, because china rose changed colour at all
- (B) neutral, because phenolphthalein and turmeric did not change
- (C) contradictory, because four indicators were used
- (D) an acid, because every result is consistent with acidity

10. A factory's effluent turns red litmus blue. The engineer must make it safe for a river before release. The single correct treatment, with its reason, is:

- (A) add more base, because base plus base cancels out
- (B) add slaked lime, because lime neutralises any effluent
- (C) add a mild acid, because the effluent is basic and an acid neutralises it
- (D) add common salt, because salt soaks up the basicity

11. A spoon of baking soda is dropped into vinegar and fizzes vigorously; the same spoon dropped into lime water gives almost no fizz. The best explanation is:

- (A) vinegar is a base and lime water is an acid, so only vinegar reacts
- (B) baking soda is an acid that fizzes only with the acidic lime water
- (C) the fizz is trapped air leaving the powder, the same in both liquids
- (D) baking soda (a base) reacts with the acid in vinegar to release a gas, but lime water is itself basic so there is little acid for it to react with

12. A pinch of crushed chalk (calcium carbonate) is added to three beakers: dilute hydrochloric acid, sodium hydroxide solution and sugar water. Fizzing is seen in:

- (A) only the sodium hydroxide beaker, where the base reacts with chalk
- (B) only the dilute hydrochloric acid beaker, where the carbonate reacts with acid to release a gas
- (C) all three beakers equally
- (D) none of them, because chalk is unreactive

13. Phenolphthalein is added to four liquids: soda water, milk of magnesia, sugar solution and washing soda solution. The number that turn it pink is:

- (A) 1
- (B) 3
- (C) 2
- (D) 0

14. A solution turns red litmus blue and is then mixed drop by drop with dilute hydrochloric acid. After many drops both litmus papers stop changing, and a few more

drops are added. Predicting the litmus result at the 'stop' point and after the extra drops gives:

- (A) first red litmus turns blue, then no change
- (B) no change in both cases, because mixing cancels everything permanently
- (C) first no change in either litmus, then blue litmus turns red
- (D) first no change, then red litmus turns blue

15. A drop of china rose indicator turns magenta in cup X and green in cup Y. You then add phenolphthalein to fresh portions of X and Y. The phenolphthalein colours are:

- (A) colourless in X and pink in Y
- (B) pink in X and colourless in Y
- (C) pink in both
- (D) colourless in both

16. A gardener has one bed too acidic for its crop and a second bed already too basic. The only treatment on hand is slaked lime. Spreading it on both beds would:

- (A) help the acidic bed by neutralising it but harm the basic bed by making it more basic
- (B) help both beds, because lime is neutral and balances any soil
- (C) harm the acidic bed and help the basic bed, because lime is acidic
- (D) have no effect on either, because lime is just a salt

17. Ammonia solution (in some glass cleaners) turns red litmus blue. Predicting what it does to turmeric paper, to china rose and to phenolphthalein gives:

- (A) turmeric turns red, china rose turns green, phenolphthalein turns pink
- (B) turmeric stays yellow, china rose turns magenta, phenolphthalein stays colourless
- (C) turmeric turns red, china rose turns magenta, phenolphthalein turns pink
- (D) turmeric stays yellow, china rose turns green, phenolphthalein stays colourless

18. A monument team is told acid rain is corroding a marble (calcium carbonate) wall. They propose washing the wall periodically with a very dilute basic solution. The chemical sense of this plan is:

- (A) unsound, because adding a base to the basic marble speeds the damage
- (B) unsound, because only more acid can clean acid off stone
- (C) pointless, because marble is acidic and unaffected by any wash
- (D) sound, because a base can neutralise the acidic residue left on the marble by acid rain

19. A liquid is reported to turn turmeric paper red yet leave both red and blue litmus papers completely unchanged. The most reasonable conclusion is:

- (A) the liquid is a special neutral substance, since litmus did not change
- (B) the report is inconsistent, because anything basic enough to turn turmeric red would also turn red litmus blue
- (C) the liquid is an acid, because turmeric changed colour
- (D) the liquid is salt water, which behaves this way

20. A chemist neutralises sulphuric acid with potassium hydroxide. Besides water, the product is best described as:

- (A) common salt (sodium chloride), as in every neutralisation
- (B) more sulphuric acid, regenerated by the reaction
- (C) a salt formed from the metal part of the base and the acid part of the acid (here potassium sulphate)
- (D) hydrogen gas together with the leftover base

21. Solution 1 turns phenolphthalein pink. Solution 2 leaves phenolphthalein colourless but turns blue litmus red. Solution 3 leaves phenolphthalein colourless and leaves both litmus papers unchanged. The families of 1, 2 and 3 are:

- (A) base, acid, neutral
- (B) acid, base, neutral
- (C) base, neutral, acid
- (D) neutral, acid, base

22. A bottle of antacid syrup is a mild base. Tracing what happens when a spoonful meets the excess hydrochloric acid in an over-acidic stomach gives the chain:

- (A) the mild base adds to the acid, making the stomach more acidic and easing the burn
- (B) the mild base neutralises the excess acid, forming a salt and water, so the burning eases
- (C) the syrup coats the stomach with sugar so the acid cannot be tasted
- (D) the syrup turns the stomach acid into a stronger acid that the body expels

23. A liquid turns blue litmus red. Predicting what red litmus, phenolphthalein, turmeric and china rose each do with the same liquid gives:

- (A) red litmus turns blue, phenolphthalein turns pink, turmeric turns red, china rose turns green
- (B) red litmus stays red, phenolphthalein stays colourless, turmeric stays yellow, china rose turns magenta
- (C) red litmus stays red, phenolphthalein turns pink, turmeric turns red, china rose turns magenta
- (D) red litmus turns blue, phenolphthalein stays colourless, turmeric turns red, china rose turns green

24. Lemon juice is added drop by drop to soap solution containing phenolphthalein, currently pink, until the pink just disappears; then more lemon juice is added and the mixture is tested with china rose. The china rose now shows:

- (A) green, because soap solution is always basic
- (B) no colour, because china rose only works on neutral liquids
- (C) magenta, but only because soap was originally a base
- (D) magenta, because the excess acid has made the mixture acidic

25. Four substances are sorted: curd, washing soda solution, common salt water and milk of magnesia. Grouped as (acids) / (bases) / (neutral), the correct grouping is:

- (A) curd and common salt water / washing soda solution / milk of magnesia
- (B) washing soda solution / curd and milk of magnesia / common salt water
- (C) curd / washing soda solution and milk of magnesia / common salt water
- (D) milk of magnesia / curd / washing soda solution and common salt water

26. A liquid turns china rose green. Asked whether it could be lemon juice, vinegar, milk of magnesia or sugar water, the only one that fits is:

- (A) lemon juice
- (B) milk of magnesia
- (C) vinegar
- (D) sugar water

27. A student neutralises an acid with a base and, just as both litmus papers stop changing, declares the endpoint reached. The single most reliable confirmation that the endpoint was reached without overshooting is that:

- (A) blue litmus stays blue and red litmus stays red, showing no excess of either acid or base
- (B) the mixture turns blue litmus red, showing the base is fully used up
- (C) the mixture turns red litmus blue, showing the acid is fully used up
- (D) the mixture turns turmeric red, the colour of a perfect neutral point

28. A fizzy soft drink contains dissolved carbon dioxide, forming carbonic acid. Predicting how it affects blue litmus, china rose and turmeric gives:

- (A) blue litmus turns red, china rose turns green, turmeric turns red
- (B) blue litmus turns red, china rose turns magenta, turmeric stays yellow
- (C) blue litmus stays blue, china rose turns magenta, turmeric turns red
- (D) blue litmus turns red, china rose turns magenta, turmeric turns red

29. A juice is reported to taste far more sharply sour than a lemon. A student argues this extreme sourness means it has become a base. The correct judgement is:

- (A) correct; very strong sourness flips a liquid into a base
- (B) wrong; the strong sourness means it is now neutral
- (C) correct; only mildly sour things are acids
- (D) wrong; a stronger sour taste is still an acid property, and greater intensity does not change the family

30. Lemon juice is added drop by drop to baking soda solution. Bubbling occurs and then stops, and at that moment neither litmus paper changes. Three more drops of lemon juice are added. The mixture is now:

- (A) still exactly neutral, because the reaction is finished
- (B) slightly acidic, because the extra acid is in excess after neutralisation was complete
- (C) basic, because baking soda is always the stronger one
- (D) unchanged, because lemon juice cannot affect a neutral mixture

31. A teacher claims: 'Neutralising hydrochloric acid with sodium hydroxide gives common salt, so neutralising any acid with any base also gives common salt.' The correct response is:

- (A) wrong; only that particular pair gives sodium chloride, while other acid-base pairs give different salts
- (B) correct; every neutralisation gives common salt and water
- (C) correct; the salt is always common salt but the water differs
- (D) wrong; neutralisation never makes common salt at all

32. Why can relying on a single phenolphthalein test mislead, so that a second indicator is often needed?

- (A) phenolphthalein turns the same pink in acids and bases, so it cannot tell them apart
- (B) phenolphthalein turns red in acids and red in bases, giving no information
- (C) phenolphthalein changes colour only in neutral solutions, never in acids or bases
- (D) phenolphthalein is colourless in both acids and neutral solutions, so a colourless result alone cannot separate the two

33. A natural indicator and a synthetic indicator are to be paired so each entry is one of each kind. The correct pairing is:

- (A) litmus (natural) and turmeric (synthetic)
- (B) phenolphthalein (natural) and china rose (synthetic)
- (C) china rose (natural) and phenolphthalein (synthetic)
- (D) turmeric (natural) and litmus (synthetic)

34. A pond used for fish has turned acidic after acidic runoff. A worker must add a base. The choice that is both a base and a cheap, commonly used one for this purpose is:

- (A) vinegar
- (B) lemon juice
- (C) slaked lime
- (D) more of the acidic runoff

35. Three liquids X, Y, Z: X turns blue litmus red, Y turns red litmus blue, Z changes neither. Treated with methyl orange (red in acids, yellow in bases), the liquid that turns it red is:

- (A) Y
- (B) X
- (C) Z
- (D) both X and Y

36. A liquid feels soapy, turns turmeric paper red, turns phenolphthalein pink and turns china rose green, but the tester refuses to taste it. The safest single classification is:

- (A) an acid, because soapy feel is an acid property
- (B) neutral, because four different tests can never all agree
- (C) impossible to classify without tasting it
- (D) a base, since soapy feel, turmeric red, phenolphthalein pink and china rose green all point the same way

37. A solution leaves phenolphthalein colourless and also leaves turmeric yellow. A student insists this proves it is an acid. The most accurate correction is:

- (A) correct; colourless phenolphthalein and yellow turmeric always mean an acid
- (B) wrong; colourless phenolphthalein and yellow turmeric always mean a base
- (C) wrong; these results always mean the solution is neutral
- (D) these results only rule out a base; the solution could be an acid or neutral, so a litmus test is needed to decide

38. A student lists tests of an unknown liquid: it tastes bitter, feels soapy, turns red litmus blue, turns methyl orange yellow and turns china rose green. The single best conclusion is:

- (A) the liquid is an acid, because methyl orange changed at all
- (B) the liquid is neutral, because so many tests agree
- (C) the results conflict, so no conclusion is possible
- (D) the liquid is a base, since every result is a base property

39. Orange juice, washing soda solution and sugar water are each tested with blue litmus, then with china rose. The substance that turns blue litmus red AND turns china rose magenta is:

- | | |
|---------------------------|------------------|
| (A) washing soda solution | (B) sugar water |
| (C) orange juice | (D) none of them |

40. Pure water is described as neutral. The single cleanest piece of evidence that distinguishes it from both acids and bases at once is that water:

- (A) turns blue litmus red, like a weak acid
- (B) leaves both red and blue litmus papers unchanged
- (C) turns red litmus blue, like a weak base
- (D) turns phenolphthalein faintly pink

41. A neutralisation is done with exactly the right amounts of acid and base, then the water is boiled off. A student claims the dry solid left behind is the leftover acid. The correct statement is:

- (A) the solid is the salt formed in the neutralisation, since no acid remains after exact neutralisation
- (B) the solid is the leftover acid, which dried out
- (C) the solid is the leftover base, which never reacts fully
- (D) there is no solid, because neutralisation removes all matter

42. A list reads: vinegar — acid, lime water — base, milk of magnesia — base, soda water — base, washing soda — base. Exactly one pairing is wrong. It is:

- | | |
|-----------------------|-------------------------|
| (A) vinegar — acid | (B) lime water — base |
| (C) soda water — base | (D) washing soda — base |

43. A liquid turns blue litmus red, but a student wants a second result to be doubly sure it is an acid. Which second result would best confirm acidity?

- 50.** A solution turns red litmus blue. It is then mixed bit by bit with dilute sulphuric acid until red litmus stays red and blue litmus stays blue. At that point the mixture is:
- (A) still basic, because the original base can never be fully cancelled
 - (B) neutral, because just enough acid has neutralised the base so neither litmus changes
 - (C) now acidic, because any added acid makes a mixture acidic
 - (D) a fresh base stronger than the first
- 51.** A student tests four liquids with red litmus: ammonia solution turns it blue, vinegar leaves it red, washing soda solution turns it blue, sugar water leaves it red. The number of bases identified is:
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
- 52.** Why is it unsafe to conclude 'this liquid is pure water' merely because it changes neither litmus paper and leaves phenolphthalein colourless?
- (A) because pure water actually turns blue litmus red, so no change means it is not water
 - (B) because only acids and bases leave litmus unchanged, never water
 - (C) because phenolphthalein turns pink in pure water
 - (D) many neutral liquids besides water, such as salt water and sugar water, give exactly these same results
- 53.** Lemon juice is added to lime water in a neutralisation. A student must say where the water in 'acid + base gives salt + water' comes from. The correct identification is:
- (A) the water is just the water the lime water was dissolved in, and no new water forms
 - (B) the water is the leftover lemon juice
 - (C) there is no water among the products, only salt
 - (D) the water is newly formed when the acid and base react, alongside the salt
- 54.** A liquid turns red litmus blue. Asked whether it could be vinegar, distilled water, lemon juice or soap solution, the only one that fits is:
- (A) soap solution
 - (B) vinegar
 - (C) distilled water
 - (D) lemon juice
- 55.** Two beakers, one of dilute acid and one of soap solution, are each touched with turmeric paper. Drops of soap solution are then stirred into the acid beaker until its turmeric paper just begins to turn red. At that final moment the acid beaker has become:
- (A) basic, since enough base was added to pass neutral and turn turmeric red
 - (B) exactly neutral, since turmeric red marks the neutral point
 - (C) still acidic, because acids are stronger than soap
 - (D) a fresh stronger acid, because mixing two liquids strengthens the acid

56. A drop of the same china rose indicator turns magenta in cup A and green in cup B. You then add turmeric paper to fresh portions of A and B. The turmeric results are:

- (A) red in A and yellow in B
- (B) yellow in A and red in B
- (C) red in both
- (D) yellow in both

57. A student claims that because milk of magnesia and lemon juice both relieve stomach discomfort in stories they've heard, both must be bases. Tracing the chemistry, the correct statement is:

- (A) both are bases, since both are taken for the stomach
- (B) both are acids, and acids always cure acidity
- (C) milk of magnesia is a base that neutralises stomach acid, but lemon juice is an acid and would not relieve an over-acidic stomach by neutralising it
- (D) milk of magnesia is an acid and lemon juice is a base

58. A solution is reported to leave blue litmus blue, leave red litmus red, and turn phenolphthalein pink. The most reasonable conclusion is:

- (A) the solution is neutral, because the litmus papers did not change
- (B) the solution is an acid, because phenolphthalein turned pink
- (C) the report is inconsistent, because anything basic enough to turn phenolphthalein pink would also turn red litmus blue
- (D) the solution is salt water, which behaves this way

59. A teacher mixes equal neutralising amounts of an acid and a base and the beaker grows warm. Asked the cause of the warmth, the best statement is:

- (A) the warmth comes from the acid evaporating, which always heats a beaker
- (B) the warmth is heat released by the neutralisation reaction itself
- (C) the warmth comes from the salt forming a stronger acid
- (D) the warmth comes from sunlight on the glass, not the reaction

60. A drop of methyl orange is added to cup R and turns red; a drop in cup S turns yellow. Phenolphthalein is then added to fresh portions of R and S. The phenolphthalein colours are:

- (A) colourless in R and pink in S
- (B) pink in R and colourless in S
- (C) pink in both
- (D) colourless in both

Solutions — Science (Class 7), Chapter 5: Acids, Bases and Salts — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	A	31	A	41	A	51	B
2	D	12	B	22	B	32	D	42	C	52	D
3	B	13	C	23	B	33	C	43	D	53	D
4	B	14	C	24	D	34	C	44	D	54	A
5	C	15	A	25	C	35	B	45	D	55	A
6	A	16	A	26	B	36	D	46	C	56	B
7	C	17	A	27	A	37	D	47	A	57	C
8	A	18	D	28	B	38	D	48	A	58	C
9	D	19	B	29	D	39	C	49	A	59	B
10	C	20	C	30	B	40	B	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Red litmus turns blue only in a base. Of the four, only sodium hydroxide is basic. Hydrochloric acid is acidic and leaves red litmus red; sugar water and common salt water are both neutral and leave red litmus unchanged. So just one strip turns blue. Salt water is neutral, not basic, so it does not count. Only one liquid is basic, so 'three' is far too many. And red litmus does change colour, but only when a base is present, so 'none' is wrong.

2. (D) Every result listed is simply the neutral result: no litmus change, colourless phenolphthalein, yellow turmeric. Many neutral liquids share this fingerprint, including sugar water and salt water, so the tests cannot single out distilled water. Distilled water, being neutral, leaves phenolphthalein colourless (not pink) and turmeric yellow (not red), so those two 'corrections' misstate how a neutral liquid behaves.

3. (B) Vinegar is acidic, so methyl orange is red. Sodium hydroxide is basic, so it is yellow. Sugar water is neutral; methyl orange is not red in a non-acid, so it shows its yellow (non-acid) colour. Thus red, yellow, yellow. Treating sugar water as acidic (red) is the trap, but

neutral solutions are not acids. Calling vinegar yellow or sodium hydroxide red simply reverses the indicator's known directions.

4. (B) The pink vanishing marks roughly the neutral point, where the base is used up. Eight further drops of an acid add acid in excess, making the mixture acidic, so blue litmus turns red. The endpoint is not 'locked'; adding more acid overshoots it. Phenolphthalein is colourless in acids, so it stays colourless rather than returning to pink. And once enough acid is present, even lime water's mixture is acidic, so 'can never become acidic' is false.

5. (C) The salt of a neutralisation is built from the metal part of the base and the acid part of the acid, so it depends on the exact pair. Potassium hydroxide with nitric acid gives potassium nitrate; sodium hydroxide with hydrochloric acid gives sodium chloride. They are different salts, so 'common salt in both' is wrong. At exact neutralisation no base is left over, and matter is not destroyed, so a solid salt remains in each.

6. (A) You neutralise each sting with the opposite family. The wasp sting is basic, so a mild acid such as vinegar relieves it. The ant sting is acidic, so a mild base such as baking soda paste relieves it. Reversing them (option with baking soda for the wasp) treats each sting with its own family and makes things worse, and using one remedy for both ignores that the two stings are opposite in family.

7. (C) P turns red litmus blue, so P is a base; china rose is green in a base, so P turns it green. Q turns blue litmus red, so Q is acidic and would turn china rose magenta. R is neutral and leaves china rose at its own colour, neither green nor magenta. Hence only P gives green. Choosing Q confuses the acid colour (magenta) with the base colour (green), and R, being neutral, does not give the base's green.

8. (A) China rose is magenta in acids, so this solution is acidic. An acid leaves red litmus red (only a base turns it blue), leaves phenolphthalein colourless (it is pink only in bases), and tastes sour. The soapy feel and red-litmus-to-blue belong to bases, not acids, and phenolphthalein does not go pink in an acid.

9. (D) Blue litmus to red, colourless phenolphthalein, yellow turmeric and magenta china rose are all the acid responses, so they agree on acid. China rose changing to magenta is precisely the acid colour, not evidence of a base. Phenolphthalein colourless and turmeric yellow happen in both acids and neutrals, so they do not prove neutral when litmus has already shown acid. The four readings are mutually consistent, not contradictory.

10. (C) Red litmus turning blue shows the effluent is basic, so it must be neutralised with an acid. Slaked lime is itself a base and would make a basic effluent more basic, so it is the wrong choice here (it is the right one only for acidic waste). Adding more base never neutralises a base, and common salt is a neutral salt that does not neutralise anything.

11. (D) The fizz comes from a base reacting with an acid to release gas. Vinegar is the acid, so baking soda fizzes in it. Lime water is basic, so there is no acid present to drive the gas-

releasing reaction, hence little fizz. Vinegar is acidic and lime water basic (not the reverse), baking soda is a base not an acid, and a real gas-releasing reaction differs from trapped air, which would behave the same in both.

12. (B) A carbonate fizzes when it meets an acid, releasing a gas. Of the three liquids only hydrochloric acid is acidic, so fizzing appears there alone. Sodium hydroxide is a base and sugar water is neutral; neither supplies the acid needed for the gas, so no fizz. Chalk is not unreactive, but it fizzes only with the acid, not with all three.

13. (C) Phenolphthalein turns pink only in bases. Milk of magnesia and washing soda solution are both basic, so two turn it pink. Soda water is acidic (carbonic acid) and sugar solution is neutral, so phenolphthalein stays colourless in those. Hence exactly two turn pink, not one, three or zero.

14. (C) At the stop point just enough acid has neutralised the base, so the mixture is neutral and neither litmus changes. The extra drops of acid push it into the acidic range, so blue litmus turns red. Mixing does not 'lock' the mixture, so a change after the extra drops is expected. The overshoot is acidic (blue-to-red), not basic, so 'red litmus turns blue' is wrong.

15. (A) China rose magenta means X is acidic; china rose green means Y is basic. Phenolphthalein is colourless in an acid and pink in a base, so X stays colourless and Y turns pink. Saying X is pink and Y colourless reverses the families, and the 'both' options ignore that the two cups are opposite in family.

16. (A) Slaked lime is a base. Adding a base to acidic soil neutralises the acidity and helps, but adding it to soil that is already basic only increases the basicity and harms the crop. Lime is not neutral, so it does not 'balance any soil'; it is a base, not an acid; and it is not an inert salt, since it actively neutralises acid.

17. (A) Red litmus turning blue shows ammonia is basic. In a base turmeric turns red, china rose turns green, and phenolphthalein turns pink. The acid responses (turmeric yellow, china rose magenta, phenolphthalein colourless) all belong to acids, so any option mixing those in is wrong for a base.

18. (D) The damage comes from acid in the rain attacking the marble. A dilute base neutralises that acidic residue, which is helpful. The base reacts with the leftover acid, not with the marble itself, so it does not speed the damage. Adding more acid would worsen the attack, and marble is basic (a carbonate), not acidic, so the last option is also wrong.

19. (B) Turmeric turns red only in a base, and a base also turns red litmus blue. So a genuine base cannot leave litmus unchanged; the two reported results contradict each other. A neutral substance leaves turmeric yellow, so turmeric-red rules out neutral. Turmeric red is a base sign, not an acid sign, and salt water is neutral and leaves turmeric yellow.

20. (C) Every neutralisation makes a salt and water, and the salt is built from the base's metal and the acid's acid part, so potassium hydroxide with sulphuric acid gives potassium sulphate. There is no sodium here, so it cannot be sodium chloride. The acid is consumed, not regenerated, and the products are salt and water, not hydrogen gas plus leftover base.

21. (A) Pink phenolphthalein means solution 1 is a base. Solution 2 has colourless phenolphthalein and turns blue litmus red, the acid signs, so it is an acid. Solution 3 shows no change anywhere, the neutral fingerprint. So base, acid, neutral. Swapping acid and base, or misreading the neutral case, gives the other options.

22. (B) An antacid works by neutralisation: the mild base reacts with the excess hydrochloric acid to give a salt and water, lowering the acidity and easing the burn. Adding a base does not make the stomach more acidic; it reduces the acid. The relief is chemical neutralisation, not a sugar coating, and certainly not the making of a stronger acid.

23. (B) Blue litmus to red means the liquid is acidic. For an acid: red litmus is unchanged, phenolphthalein stays colourless, turmeric stays yellow, and china rose turns magenta. The base responses (red litmus to blue, phenolphthalein pink, turmeric red, china rose green) do not apply to an acid, so any option mixing those in is wrong.

24. (D) The pink vanishing marks the neutral point; further lemon juice makes the mixture acidic. China rose is magenta in an acid, so it shows magenta. It does not stay green, because the mixture is no longer basic. China rose does change colour in acids and bases (its green response is for bases), so 'only works on neutral liquids' is false, and the magenta is caused by the present acidity, not by soap's original family.

25. (C) Curd is acidic (lactic acid). Washing soda solution and milk of magnesia are both bases. Common salt water is neutral. So acids = curd, bases = washing soda and milk of magnesia, neutral = salt water. Placing salt water as an acid, or calling washing soda an acid, or calling milk of magnesia an acid, all misassign a family.

26. (B) China rose is green only in a base. Of the four, only milk of magnesia is a base. Lemon juice and vinegar are acids (china rose would be magenta), and sugar water is neutral (china rose keeps its own colour). So milk of magnesia is the only fit.

27. (A) At exact neutral, no acid or base is in excess, so neither litmus changes: blue stays blue and red stays red. If blue litmus turned red the mixture would be acidic (overshot with acid); if red litmus turned blue it would be basic (overshot with base). Turmeric red is the base colour, not a neutral marker, so it would actually signal an excess of base.

28. (B) Carbonic acid makes the drink acidic. In an acid blue litmus turns red, china rose turns magenta, and turmeric stays yellow (it turns red only in bases). Options that put china rose green or turmeric red are using base responses, and 'blue litmus stays blue' contradicts the drink being acidic.

29. (D) Sourness is a property of acids, and a more intense sour taste simply means a more strongly acidic liquid; intensity does not switch the chemical family. So the juice remains an acid, not a base and not neutral. Mild and strong sour substances are both acids, so 'only mildly sour things are acids' is also wrong.

30. (B) Bubbling stopping with no litmus change means the acid and base have neutralised each other and the mixture is neutral. Three further drops of lemon juice add acid in excess, so the mixture becomes slightly acidic. It is not 'locked' at neutral, baking soda is not automatically the stronger one, and an acid certainly can change a neutral mixture by making it acidic.

31. (A) The salt formed depends on the specific acid and base, drawing the metal from the base and the acid part from the acid. Hydrochloric acid with sodium hydroxide happens to give sodium chloride (common salt), but, say, nitric acid with potassium hydroxide gives potassium nitrate. So 'always common salt' is wrong, while 'never common salt' is also wrong because that one pair does make it.

32. (D) Phenolphthalein is pink only in a base and colourless in everything else, so a colourless reading could be an acid or a neutral solution; a litmus test is needed to decide between them. It is not the same colour in acids and bases (it is pink in bases, colourless in acids), it does not turn red, and it does change in bases, so the other statements misdescribe it.

33. (C) Litmus, turmeric and china rose are all natural (from a lichen and from plants); phenolphthalein is the synthetic, lab-made one. So a valid natural-plus-synthetic pair is china rose with phenolphthalein. Turmeric is natural (not synthetic), litmus is natural (not synthetic), and phenolphthalein is synthetic (not natural), so the other pairings mislabel one member.

34. (C) An acidic pond is neutralised by adding a base, and slaked lime is the cheap, widely used base for this. Vinegar and lemon juice are acids, which would worsen the acidity, and adding more acidic runoff is exactly the wrong direction.

35. (B) Methyl orange is red only in an acid. X turns blue litmus red, so X is the acid, and it turns methyl orange red. Y is a base (red litmus to blue) and would give yellow; Z is neutral and gives the non-acid (yellow) colour. So only X turns methyl orange red, not Y or both.

36. (D) Soapy feel, turmeric red, phenolphthalein pink and china rose green are all base properties, and they agree, so the liquid is a base. Soapy feel is a base property, not an acid one. Four tests can agree, and here they do, so 'neutral because they disagree' is false. Tasting is unnecessary and unsafe; the indicators already settle the family.

37. (D) Phenolphthalein is colourless and turmeric yellow in both acids and neutral solutions, so together they only show the solution is not a base. To tell an acid from a

neutral solution you must use litmus. Hence the student cannot conclude 'acid' from these two alone; nor do they prove 'base' or 'neutral' specifically.

38. (D) Bitter taste, soapy feel, red litmus to blue, methyl orange yellow and china rose green are all base properties, and they agree, so the liquid is a base. Methyl orange yellow is the base colour (red is the acid colour), so it does not point to acid. The agreement of many tests supports 'base', not 'neutral', and the results are consistent rather than conflicting.

39. (C) Turning blue litmus red and turning china rose magenta are both acid responses, so the substance that gives both must be acidic. Orange juice is the only acid of the three, so it turns blue litmus red and china rose magenta. Washing soda solution is a base, so blue litmus would stay blue and china rose would turn green. Sugar water is neutral, so blue litmus stays blue and china rose keeps its own colour. Only orange juice gives both acid results.

40. (B) Leaving both litmus papers unchanged is exactly the neutral result and sets water apart from acids (which redden blue litmus) and bases (which blue red litmus) in one test. Turning blue litmus red would make it an acid; turning red litmus blue would make it a base; and phenolphthalein pink would make it a base too, none of which is true for neutral water.

41. (A) At exact neutralisation the acid and base are fully consumed, producing a salt and water. Boiling off the water leaves that salt as a solid. No acid is left to dry out, and the base is fully reacted too, so neither leftover survives. Matter is not destroyed, so a solid salt does remain.

42. (C) Soda water contains carbonic acid and is therefore acidic, so calling it a base is the wrong pairing. Vinegar is correctly an acid, while lime water, milk of magnesia and washing soda are all correctly bases. Only the soda water entry is mislabelled.

43. (D) China rose turning magenta is an acid response, so it confirms acidity in agreement with the litmus result. Turmeric red and phenolphthalein pink are base responses, which would contradict an acid; and red litmus turning blue is also a base sign. So only the china rose magenta result properly backs up 'acid'.

44. (D) China rose is magenta in acids and green in bases, so cup P (magenta) is the acid and cup Q (green) is the base. Magenta marks an acid, not a base, so saying P is a base is wrong, and green marks a base, not an acid. China rose does distinguish acids from bases, so 'neither' is also wrong.

45. (D) Methyl orange is yellow only when the solution is no longer acidic, i.e. once a base is present in excess. So yellow means the acid has been neutralised and the mixture has moved into the basic range. Methyl orange is not specifically the 'neutral' colour; an acid

does not turn into a base; and the indicator has not run out, it has simply changed colour as designed.

46. (C) Lime water and washing soda solution are bases (two). Curd and tamarind water are acids, and milk is roughly neutral when fresh. So exactly two are bases, not one, three or four.

47. (A) Sugar is neutral and does not neutralise an acid; relief from an acidic sting needs a base such as baking soda paste. Sugar is not a base, sweetness has nothing to do with neutralising acids, and the bee sting is acidic (not basic), so treating it as basic is doubly wrong.

48. (A) An antacid is a base that reacts with the excess stomach acid to form a salt and water, which is precisely acid + base giving salt + water. Dissolving sugar is just mixing, an apple browning is a different chemical change, and ice melting is only a change of state, so none of those is a neutralisation.

49. (A) Turmeric turns red only in a base, so the solution is basic. In a base blue litmus is unchanged (it reddens only in acid), phenolphthalein turns pink, and methyl orange turns yellow. The acid responses (blue litmus to red, phenolphthalein colourless, methyl orange red) do not apply to a base, so the options containing those are wrong.

50. (B) When both litmus papers stop changing, no excess acid or base remains, so the mixture is neutral. A base can be fully neutralised by enough acid, so 'never cancelled' is wrong. Adding acid up to but not beyond the endpoint gives neutral, not automatically acidic, and neutralising a base does not create a stronger base.

51. (B) Red litmus turns blue only in a base. Ammonia solution and washing soda solution both turn it blue, so two bases. Vinegar is an acid and sugar water is neutral; both leave red litmus red. Hence exactly two bases.

52. (D) No litmus change plus colourless phenolphthalein is simply the neutral fingerprint, which salt water and sugar water share with pure water, so the tests cannot prove the liquid is water specifically. Pure water does not redden blue litmus, it does leave litmus unchanged (so 'never water' is wrong), and phenolphthalein stays colourless in neutral water rather than turning pink.

53. (D) In neutralisation the acid and base react to produce both a salt and fresh water; this water is a genuine product of the reaction, not the solvent that was already present and not leftover lemon juice. Saying there is 'no water' contradicts the very equation acid + base gives salt + water.

54. (A) Red litmus turns blue only in a base, and of the four only soap solution is basic. Vinegar and lemon juice are acids, which leave red litmus red, and distilled water is neutral, which also leaves it unchanged. So soap solution is the only fit.

55. (A) Turmeric turns red only in a base, so the moment its paper reddens the mixture has become basic, meaning enough soap solution (a base) was added to overshoot neutral. Turmeric red is not the neutral marker (neutral leaves turmeric yellow), the mixture is no longer acidic, and adding a base does not make a stronger acid.

56. (B) China rose magenta means A is acidic; china rose green means B is basic. Turmeric stays yellow in an acid and turns red in a base, so A gives yellow and B gives red. Reversing them, or claiming both are the same colour, ignores that A and B are opposite families and that turmeric reddens only for the base.

57. (C) An over-acidic stomach is eased by a base, and milk of magnesia is a base that neutralises the excess acid. Lemon juice, however, is an acid and would not neutralise stomach acid; it would add to the acidity. So they are not both bases, not both acids, and the families are not the other way round (milk of magnesia is the base, lemon juice the acid).

58. (C) Phenolphthalein turns pink only in a base, and a base also turns red litmus blue. So red litmus staying red while phenolphthalein is pink is a contradiction. A neutral solution would leave phenolphthalein colourless, not pink, so the no-change litmus does not make it neutral here. Pink phenolphthalein is a base sign, not an acid sign, and salt water (neutral) would leave phenolphthalein colourless.

59. (B) Neutralisation releases heat, so the beaker warms as the acid and base react. The warmth is not from the acid evaporating (evaporation would cool, not heat), nor from any 'stronger acid' (neutralisation consumes the acid), nor from sunlight, since the warming tracks the reaction itself.

60. (A) Methyl orange is red in acids and yellow in bases, so R is acidic and S is basic. Phenolphthalein is colourless in an acid and pink in a base, so R stays colourless and S turns pink. Reversing them confuses the families, and the 'both' options ignore that R and S are opposite in family.